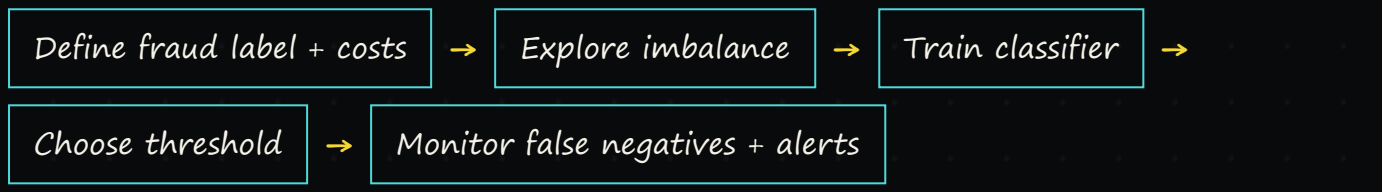
Using AutoML to Detect Fraudulent Transactions

Develop a fraud classifier while accounting for rare positives, threshold-dependent costs, feature leakage, and operational trade-offs.

Original study notes based on Gwendolyn Stripling and Michael Abel, First edition, 2023. Book pp. 113–146. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.



Detailed Notes

Use case, dataset, and preprocessing

Source: book pp. 113–131 / PDF p. 129

Fraud detection is a binary classification problem, but fraudulent transactions may be a tiny fraction. Explore class counts, transaction types, balance changes, amount ranges, and suspicious patterns. Preserve the target label and remove identifiers or post-event fields that would leak the answer.

Use a time-aware split if deployment predicts future transactions; random splits can leak repeated actors or temporal patterns. Visualizations can expose relationships and anomalies but cannot establish fraud causation. Save cleaned data and document transformations.

Classification models and metrics

Source: book pp. 132–134 / PDF p. 148

For a confusion matrix, track true/false positives and negatives at the operational threshold. Accuracy can be misleading: predicting "legitimate" for every transaction may appear strong under severe imbalance.

> $Precision = TP/(TP+FP)$ $Recall = TP/(TP+FN)$
> $F1 = 2 \times Precision \times Recall/(Precision+Recall)$

Precision measures alert reliability; recall measures detected fraud. ROC-AUC summarizes ranking over thresholds; PR-AUC is often informative for rare positives. Choose a threshold using fraud costs, investigation capacity, customer friction, and acceptable missed-loss rates.

AutoML training, evaluation, and attribution

Source: book pp. 134–141 / PDF p. 150

Create a managed classification dataset, identify the positive class carefully, train, and inspect class-specific metrics. Compare confusion matrices and threshold curves rather than selecting the highest headline accuracy. Check whether the validation split represents future transactions.

Feature importance can reveal associations used by the model, not justify blocking a customer or prove a feature causes fraud. Review protected/sensitive proxies, false-positive cohorts, and model explanations with responsible stakeholders.

Predictions and operational safeguards

Source: book pp. 142–146 / PDF p. 158

Deploy a prediction endpoint only after an acceptance threshold is met. Test request schema, latency, throughput, access controls, and fallback behavior. A probability score can support a queue or step-up review; it need not automatically reject a transaction.

Track false negatives from later investigations, false positives/appeals, drift in transaction patterns, and serving cost. Undeploy idle managed endpoints when they incur ongoing charges. Retrain only with reliable adjudicated labels and leakage-safe temporal data.

Revision Check

1. Why can accuracy be useless on an imbalanced fraud set?
2. What costs drive the threshold?
3. How might time leakage occur?
4. Should a score automatically block a transaction?

Fraud detection is an operating policy, not just a classifier. Thresholds, investigations, customer harm, and feedback labels all shape success.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.